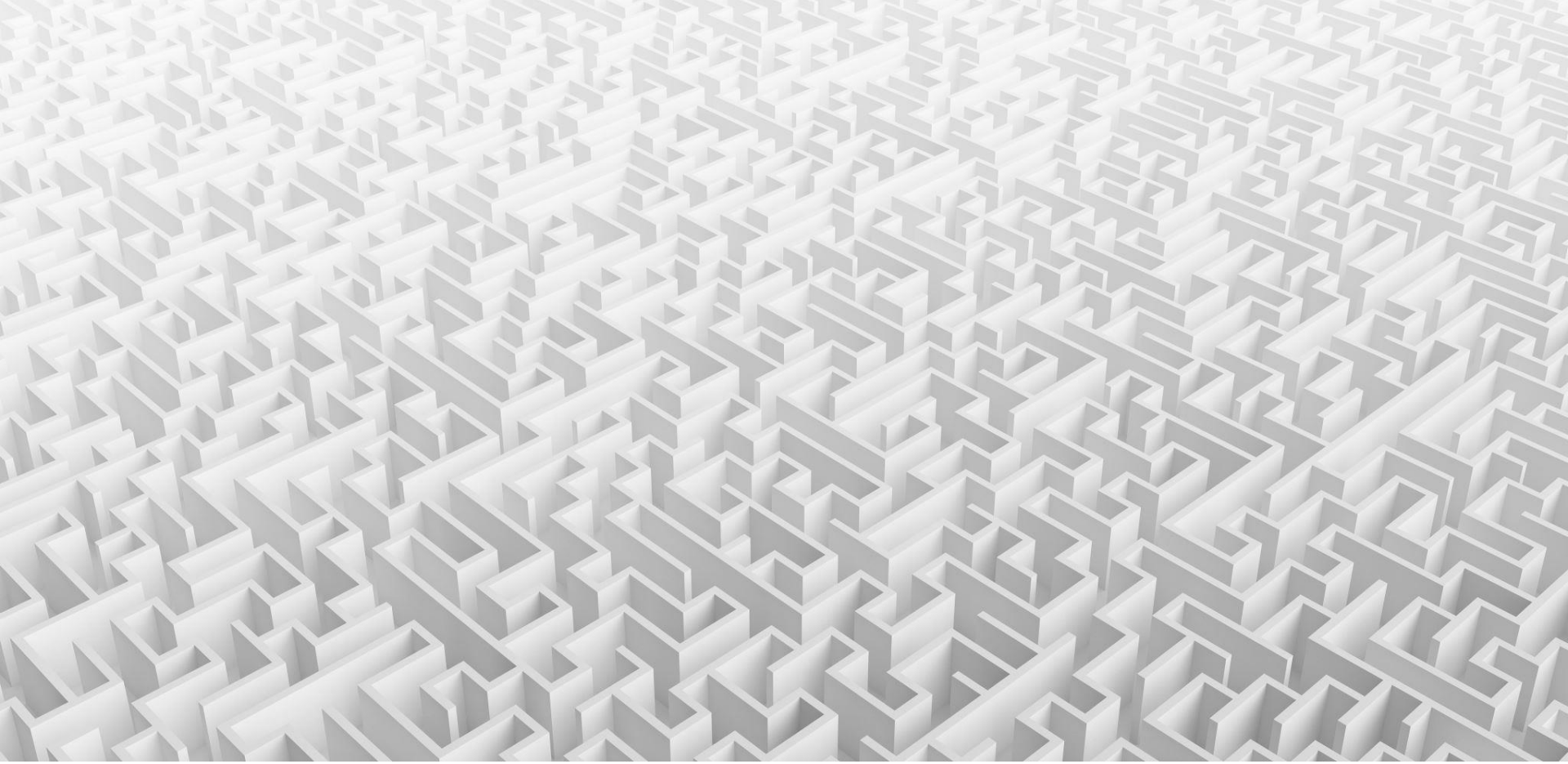

Practical Labs

Report structure and datasets

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Report structure

1. Describe the **biological problem** and the objective of the study;

2. Describe the **experimental design** *(wet lab protocol, library size, il the Material and methods section of the paper is MIAMI compliant)*

4. Il report must include, as a **minimum requirement**, the implementation of the following steps: exploratory data analysis, normalization, identification of differentially expressed genes (DEGs), and gene set analysis;

5. The use of at least one **new Bioconductor package** implementing analyses similar to those presented during the practical sessions will be **considered added value in the evaluation of the project work**. The inclusion of these additional analyses also requires that you understand them and explain both the methodology and its rationale in the report and during the presentation.

7. **Biological interpretation** of the results and if possible comparison with the results of the original paper;

Report structure

5. The use of at least one **new Bioconductor package** implementing analyses similar to those presented during the practical sessions will be **considered added value in the evaluation of the project work**. The inclusion of these additional analyses also requires that you understand them and explain both the methodology and its rationale in the report and during the presentation.

A list of possible alternative packages to use (although you do not necessarily need to limit yourselves to these):

DESeq2
Limma
NOISeq
enrichr
GSVA
Sva (function combat)
RUVSeq
bluster
xCell2
Signifinder
simplifyEnrichment
…

Preparation of the material to be sent

1. **Before May 25** you must upload in a dedicated session of the moodle:

   - the R code as an Rmd file
   - the compiled HTML/PDF file with the final results,
   - We will notify via moodle when the submission is open.

1. The **rmd file should be reproducible**, **the code should run properly** without errors.

2. For the presentation of the work, it is expected to have **a separate PDF/PPT** that summarise the entire work.

3. Each group **will have 10 to 15 minutes** for the presentation. It is expected that each member of the group has contributed to the analyses, to the preparation of the submitted documents, and will contribute to the presentation. During the presentation, the instructors will ask theoretical and practical questions related to the analyses presented.

Problems with the analyses

During the practical labs on **April 1st, 14th and 22th**, you will independently work with your dataset under our supervision.

Should you encounter any problems or errors that you're unable to resolve, you can seek **assistance during the practical labs** or reach out via **email to:**

Prof. Sales
Prof. Romualdi
CC: Sara Potente (sara.potente@phd.unipd.it)
CC: Ilaria Billato (ilaria.billato@unipd.it)
CC: Marco Rota Negroni (marco.rotanegroni@studenti.unipd.it)
CC: Elena Zuin (elena.zuin.3@studenti.unipd.it)